



Stony Brook Institute
at Anhui University

安徽大学纽约石溪学院

CSE 306 Operating Systems Command-Line Interface (CLI)

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Instructor: Hao Li

Virtual Machine - Before The Journey

- Oracle VirtualBox
 - Prepare a virtual machine (VM) based Linux workstation in Windows OS
 - First, we need to install the Oracle VirtualBox (VirtualBox 6.1.42) software
 - Second, download the Linux installation (Ubuntu LTS 18.04) installation image
 - Third, create a VM in VirtualBox and install the Linux installation image in the VM
 - Finally, perform some necessary configuration within the VM for code development.



Virtual Machine - Before The Journey

- Oracle VirtualBox
 - Download a VirtualBox setup file from its official website:
<https://www.virtualbox.org/>
 - For example, you may go to <https://www.virtualbox.org/wiki/Downloads> and clicks on “Windows hosts” (under the title “VirtualBox {VER_NUM} platform packages”) to download a VirtualBox setup file for Windows OS.
 - Once the setup file is downloaded, run it to install the VirtualBox software.



Virtual Machine - Before The Journey

- Linux installation image
 - VirtualBox is only a tool for providing a container (or virtual environment) separated from the host Windows OS, yet VirtualBox itself does not provide any OS workstation by itself
 - You need to install manually a Linux OS inside a container created by VirtualBox.
 - Download a Linux installation image
 - For example, download the Ubuntu LTS 18.04 installation image `ubuntu-18.04.x-desktop-amd64.iso`, which is a suitable Long Term Support (LTS) version of Ubuntu

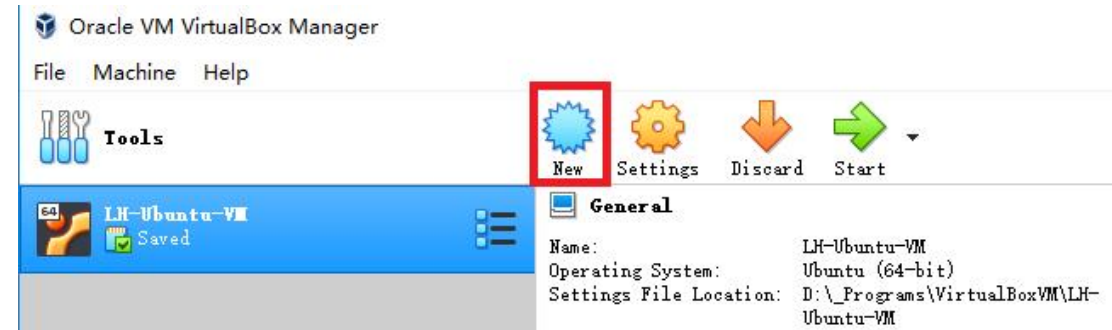
Virtual Machine - Before The Journey

- Create a VM in VirtualBox
 - Run VirtualBox and you would see a GUI somewhat as follows:
 - You see an entity named “LH-Ubuntu-VM” in the left column; this is a VM that I have already created. If you run VirtualBox for the first time, then nothing exists in the left column; your task now is right to create a new VM and install the Ubuntu installation image inside it.

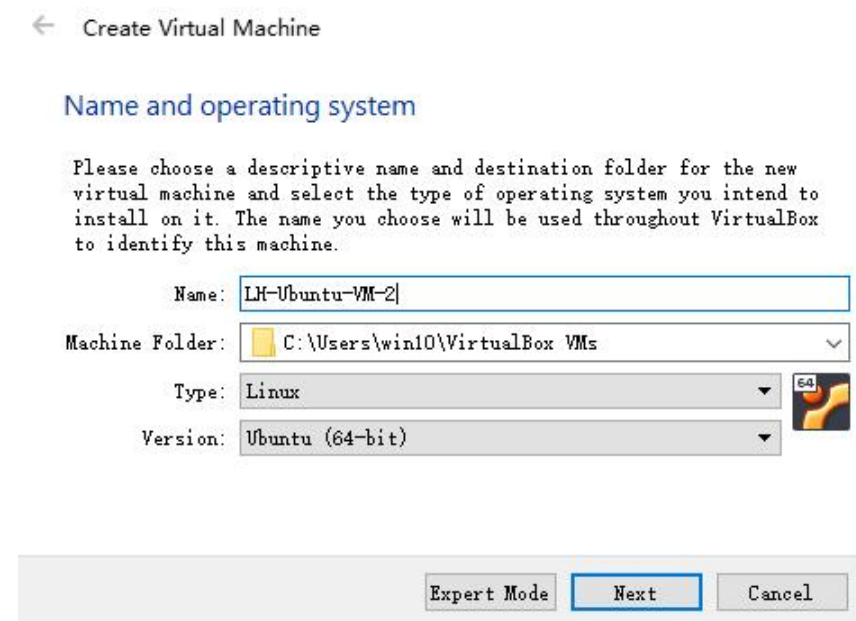


Virtual Machine - Before The Journey

- Create a VM in VirtualBox
 - 1) Click on New



- 2) Choose whatever name you prefer for your VM; set “Type” and “Version” as below



Virtual Machine - Before The Journey

- Create a VM in VirtualBox
 - 3) Set the virtual memory size according to your PC conditions
 - Of course, you should not have a virtual memory size beyond your actual PC memory size; otherwise, your PC will be overwhelmed when you run the VM
 - To install Ubuntu, we had better set a virtual memory size not too small either; a size of 2048 MB would be suitable.
 - 4) Tick “Create a virtual hard disk now” and go next
 - 5) Just choose the default “VDI (VirtualBox Disk Image)” format.

← Create Virtual Machine

Memory size

Select the amount of memory (RAM) in megabytes to be allocated to the virtual machine.

The recommended memory size is 1024 MB.



Hard disk

If you wish you can add a virtual hard disk to the new machine. You can either create a new hard disk file or select one from the list or from another location using the folder icon.

If you need a more complex storage set-up you can skip this step and make the changes to the machine settings once the machine is created

The recommended size of the hard disk is 10.00 GB.

- ☐ Do not add a virtual hard disk
- ☒ Create a virtual hard disk now
- ☐ Use an existing virtual hard disk file

LH-Ubuntu-VM.vdi (Normal, 30.00 GB)

Hard disk file type

Please choose the type of file that you would like to use for the new virtual hard disk. If you do not need to use it with other virtualization software you can leave this setting unchanged.

- ☒ VDI (VirtualBox Disk Image)
- ☐ VHD (Virtual Hard Disk)
- ☐ VMDK (Virtual Machine Disk)

Virtual Machine - Before The Journey

- Create a VM in VirtualBox
 - 6) There are two modes for VM storage, namely the “dynamically allocated” mode and the “fixed size” mode; choose the “fixed size” mode.
 - 7) Set a VM size and then click on “Create” to create the VM.
 - Once all above is done, a new VM named “LH-Ubuntu-VM-2” is created in my VirtualBox. Similarly, a VM with the name you set would also be created in your VirtualBox

Storage on physical hard disk

Please choose whether the new virtual hard disk file should grow as it is used (dynamically allocated) or if it should be created at its maximum size (fixed size).

A **dynamically allocated** hard disk file will only use space on your physical hard disk as it fills up (up to a maximum **fixed size**), although it will not shrink again automatically when space on it is freed.

A **fixed size** hard disk file may take longer to create on some systems but is often faster to use.

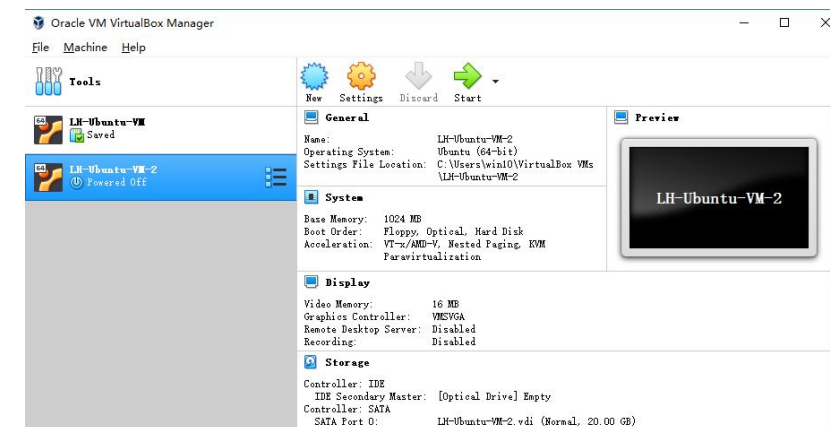
- ☐ Dynamically allocated
- ☒ Fixed size

File location and size

Please type the name of the new virtual hard disk file into the box below or click on the folder icon to select a different folder to create the file in.

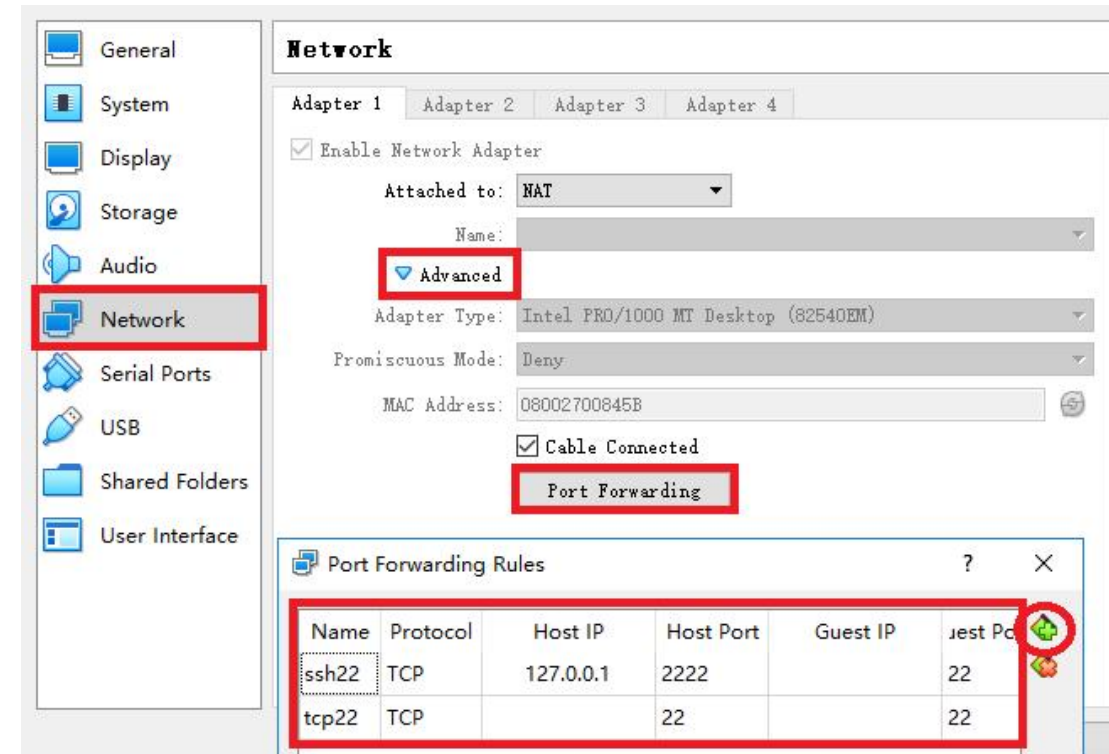
LH-Ubuntu-VM-2

Select the size of the virtual hard disk in megabytes. This size is the limit on the amount of file data that a virtual machine will be able to store on the hard disk.



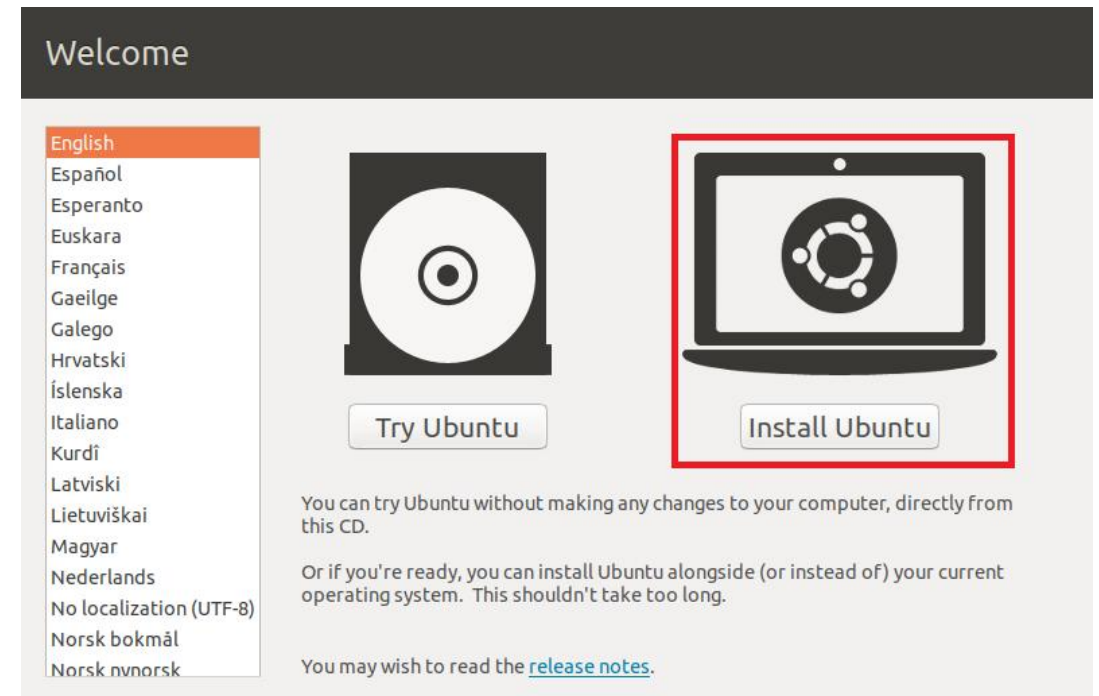
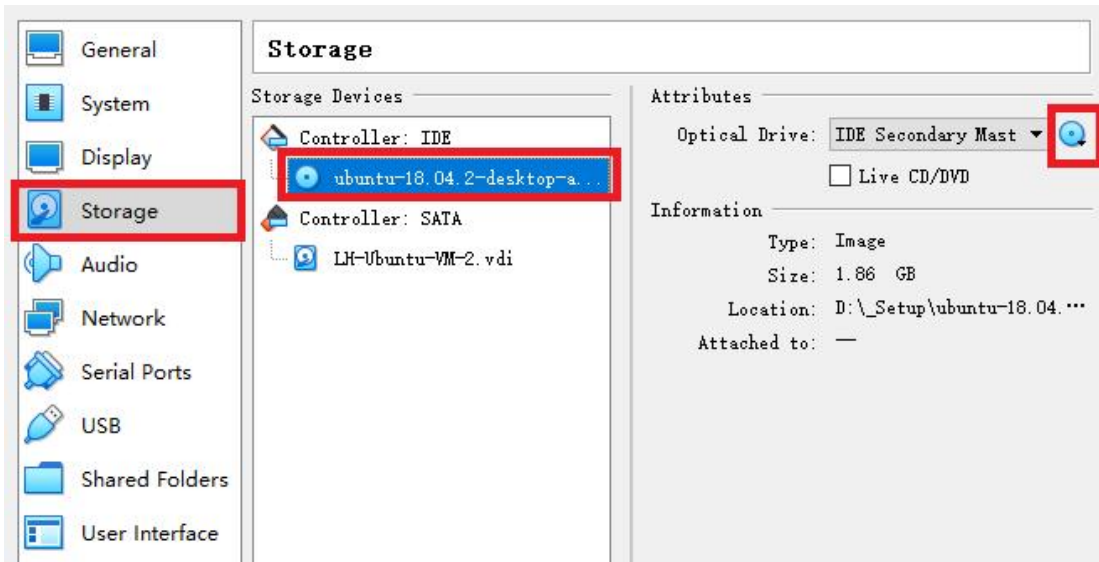
Virtual Machine - Before The Journey

- Install Ubuntu in the created VM
 - 1) Activate (i.e. click on) the created VM; click on “Settings” to make preliminary preparations
 - 2) Network configuration: Network --> Port Forwarding --> Click on “+” --> Set “Port Forwarding Rules”



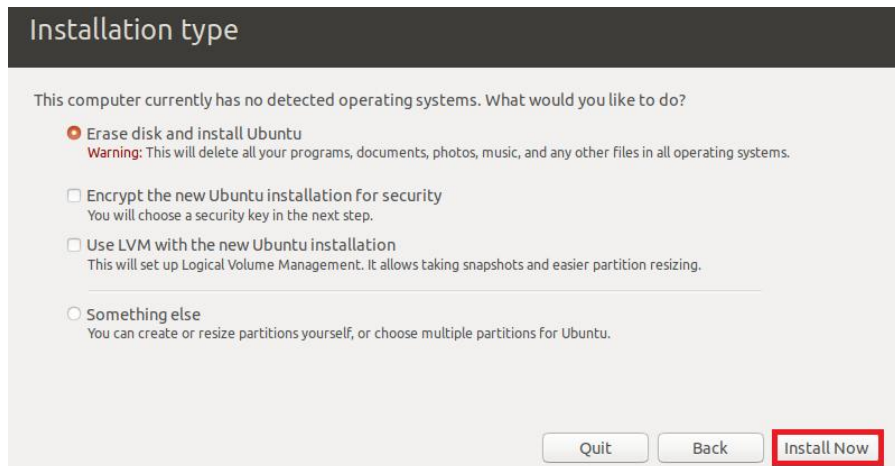
Virtual Machine - Before The Journey

- Install Ubuntu in the created VM
 - 3) Storage configuration: Storage --> Click on the small disk icon on the right side -> Choose the ubuntu-18.04.x-desktop-amd64.iso installation image you downloaded below --> Click on “OK” to quit “Settings”.
 - 4) Click on “Start” to start the installation of Ubuntu.
 - 5) Click on “Install Ubuntu”.



Virtual Machine - Before The Journey

- Install Ubuntu in the created VM
 - 5) Click on “Install Ubuntu”.
 - For the “Installation Type”, here you can just choose “Erase disk and install Ubuntu” and do not need to worry that data on your PC would be really erased, because all operations involved here are only within your VM.
 - 6) Once “Install Now” is clicked on, installation of Ubuntu in your VM will formally start. At the beginning of the installation, you will be prompted to set user names, passwords etc for your VM; once done, simply wait for the installation process to finish.



Common Commands

• ls	cd	pwd	mkdir	touch
• cp	mv	rm	cat	less
• head	tail	find	grep	ps
• kill	df	du	chmod	chown
• sudo	man	history	passwd	top
• sed	awk	ssh	scp	sshd

Common Commands

- **ls**

- List directory contents
- `ls -l`
- `ls -a`
- `ls -la` / `ls -al`
- `ls --recursive` / `ls -la --recursive`

- **mkdir**

- Make directories
- `mkdir dir1 dir2 dir3`

- **touch**

- Change file timestamps (to create files as well)
- `touch file1 file2 file3`

Common Commands

- **pwd**
 - Print (name of) working/current directory
- **cd**
 - Change (working) directory
 - `cd {an_absolute_path}`
 - `cd {a_relative_path}`
 - `cd ./ {a_relative_path}`
 - `cd ../ {a_relative_path}`
 - `cd ../../ {a_relative_path}`

Common Commands

- **rm**

- Remove files and directories
- `rm file1 file2`
- `rm -r dir1 dir2`

- **cp**

- Copy files and directories
- `cp {file_paths} {a_new_path}`
- `cp -r {dir_paths} {a_new_path}`

- **mv**

- Move (rename) files and directories